KrishiIQ Mobile Application: App Flow Document

1. Introduction

KrishiIQ is an innovative AI-driven advisory system designed to empower smallholder farmers in India. This document outlines the key user journeys, screen flows, and navigation for the KrishiIQ Flutter mobile application, focusing on the farmer's interaction with the system. The application aims to provide personalized, real-time agricultural advice by integrating various data sources and leveraging AI for predictive intelligence.

2. Key User Roles

Farmer: The primary user of the mobile application. Farmers will interact with the app to input data, receive personalized advisories, detect crop diseases, and monitor their farm's health.

3. Core Features (Farmer App)

The KrishiIQ mobile app for farmers includes the following core functionalities:

- **Data Collection**: Gathers data from IoT soil sensors, weather APIs, and direct farmer inputs (e.g., crop history, soil health card details).
- **AI Crop Disease Detection**: Analyzes crop images using EfficientNet to identify diseases and suggest remedies.
- **Personalized Advisory**: Provides AI-driven recommendations for optimal sowing times, irrigation schedules, and pest outbreak detection.
- **Multilingual Voice/Text Advisory**: Delivers advice in Tamil, Hindi, and English, catering to low-literacy farmers.
- **Offline Capability**: Ensures continuous access to essential features even without internet connectivity.
- **Carbon Footprint & Water Conservation Tracking**: Monitors and provides insights into environmental impact.

4. User Journeys and Screen Flows

This section details the typical paths a farmer will take through the KrishiIQ application.

4.1. Onboarding and Registration

Objective: Allow new farmers to create an account and set up their initial farm profile.

Step	Screen/Action	Description	Inputs/Outputs
1	**Welcome Screen**	Displays app's value proposition.	Option to 'Register' or 'Login'.
2	**Registration Form**	Collects basic farmer information.	Name, Phone Number, Preferred Language (Tamil, Hindi, English). OTP verification.
3	**Farm Profile Setup**	Gathers initial farm details.	Farm Location (GPS/Manual), Crop Type, Soil Type (if known), Land Area.
4	**Sensor Connection (Optional)**	Guides farmer to connect IoT sensors.	Instructions for pairing sensors.
5	**Onboarding Complete**	Confirmation and redirection to Dashboard.	Success message.

4.2. Dashboard/Home Screen

Objective: Provide an overview of farm status, alerts, and quick access to main features.

Section	Content	Actions
Farm Summary	Current weather, soil moisture, last advisory.	View detailed weather, soil data.
Alerts/Notifications	Pest outbreak warnings, irrigation reminders, sowing recommendations.	Tap to view details/take action.
Quick Actions	Capture Crop Image, Input Farm Data, View Advisory.	Direct access to key features.
Navigation Bar	Home, Advisory, Sensors, History, Profile.	Navigate to main sections.

4.3. Data Input Flow

Objective: Allow farmers to manually input or connect data sources.

4.3.1. Manual Farm Data Input

Step	Screen/Action	Description	Inputs/Outputs
1	**Select 'Input Farm Data'**	From Dashboard or 'History' section.	List of data categories (Crop History, Soil Health Card, etc.).
2	**Choose Data Category**	E.g., 'Crop History'.	Form fields relevant to the category.
3	**Enter Details**	Input past crop details, yield, fertilizers used.	Save button.
4	**Confirmation**	Data saved successfully.	Return to previous screen.

4.3.2. IoT Sensor Management

Step	Screen/Action	Description	Inputs/Outputs
1	**Navigate to 'Sensors'**	From Navigation Bar.	List of connected sensors, option to 'Add New Sensor'.
2	**Add New Sensor**	Guides through pairing process.	Bluetooth/Wi-Fi pairing, sensor ID input.
3	**Sensor Data View**	Real-time data from connected sensors.	Soil moisture, temperature, pH levels.

4.4. Crop Disease Detection Flow

Objective: Enable farmers to identify crop diseases using image analysis.

Step	Screen/Action	Description	Inputs/Outputs

1	**Select 'Capture Crop Image'**	From Dashboard Quick Actions.	Camera interface or 'Upload from Gallery' option.
2	**Capture/Upload Image**	Farmer takes a photo of the affected crop or selects one.	Image preview.
3	**Image Analysis**	App processes the image using AI.	Loading indicator.
4	**Diagnosis & Recommendation**	Displays identified disease and suggested remedies.	Disease name, severity, treatment steps (text/voice).

4.5. Personalized Advisory Flow

Objective: Provide timely and relevant agricultural advice.

Step	Screen/Action	Description	Inputs/Outputs
1	**Navigate to 'Advisory'**	From Navigation Bar or Dashboard Alert.	List of current and past advisories.
2	**View Specific Advisory**	Tap on an advisory (e.g., 'Sowing Recommendation').	Detailed advice (text/voice), rationale.
3	**Voice Playback**	Option to listen to the advisory.	Play/Pause button, language selection.
4	**Pest Outbreak Alert**	Special advisory for detected anomalies.	Location of outbreak, recommended actions.

4.6. Historical Data and Analytics

Objective: Allow farmers to review past performance and environmental impact.

Step	Screen/Action	Description	Inputs/Outputs
1	**Navigate to 'History'**	From Navigation Bar.	Options: 'Crop History', 'Sensor Data Logs', 'Carbon Footprint', 'Water Usage'.

2	**Select Category**	E.g., 'Carbon Footprint'.	Visualizations (charts/graphs) and summary data.
3	**Filter/Drill Down**	View data by date range, crop cycle.	Interactive charts.

4.7. Settings and Profile Management

Objective: Manage app settings and personal information.

Step	Screen/Action	Description	Inputs/Outputs
1	**Navigate to 'Profile'**	From Navigation Bar.	Options: 'Edit Profile', 'Change Language', 'Manage Sensors', 'Help & Support'.
2	**Change Language**	Select preferred language for app interface and advisories.	Tamil, Hindi, English options.
3	**Edit Profile**	Update personal details.	Name, phone number, farm location.

5. Navigation Structure

The primary navigation will be facilitated by a bottom navigation bar for quick access to core sections, complemented by in-app navigation for specific feature flows.

Bottom Navigation Bar:

- **Home**: Dashboard overview.
- **Advisory**: List of personalized recommendations and alerts.
- **Sensors**: Manage and view data from IoT sensors.
- **History**: Access past data, analytics, and environmental tracking.
- **Profile**: App settings, personal information, and help.

In-App Navigation:

- Back buttons for returning to previous screens.

- Contextual links and buttons within screens to initiate specific actions (e.g., 'Capture Image', 'Add Sensor').

6. Offline Capability Considerations

For offline functionality, the app will:

- Store critical advisory data, sensor thresholds, and disease detection models locally.
- Queue farmer inputs and sensor data for synchronization once connectivity is restored.
- Provide basic advisory and disease detection capabilities using local models.

7. Conclusion

This document provides a foundational understanding of the KrishiIQ mobile application's user experience and flow. These flows will serve as a basis for detailed UI/UX design and development, ensuring a user-friendly and effective tool for farmers.